

Monthly Intelligence Report

Edition 003 — The North West-West Midlands Seismic Corridor

Where the energy transition is outrunning the grid. edition · June 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

3,150
342 EHV · 2,808 HV

GRID LINES MAPPED

~22,765
~87,000 km coverage

MC SIMULATIONS

35.3 M
10,000 per substation

PUBLIC DATA SOURCES

30+
95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 65,300+ source data points per run, executes 35.3 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 794 million arithmetic operations — producing 209,722 output data points with full uncertainty quantification across 3,150 substations and ~22,765 grid lines spanning ~105,000 km.

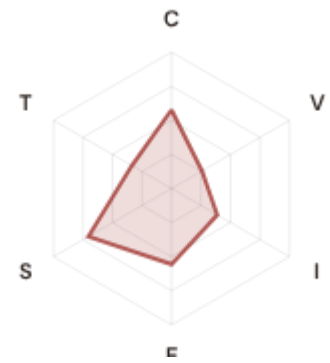
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering the UK's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Grid Supply Point Down 3106

Down, Northern Ireland · 33 kV

0.552 **High** **0.180** **93th**
R_FINAL CLASSIFICATION CI WIDTH FLEET PERCENTILE



COMPONENTS

C Continuity: 0.577 / 0.30 (192%) ▲
I Infrastructure: 0.387 / 0.25 (155%) ▲
S Saturation: 0.704 / 0.20 (352%)
V Voltage: 0.258 / 0.10 (258%)
E Economic: 0.556 / 0.10 (556%)
T Transition: 0.336 / 0.05 (671%)

MODIFIERS

R3 Consequence: 1.147 ↑ amplifies
R4 Graph Criticality: 0.956 ↓ dampens
R6a Restoration: 0.995 → neutral
R6b Seismic: 1.000 → neutral
R7 Digital Readiness: 0.990 → neutral

This month's spotlight — automatically selected for June 2026 — is **Grid Supply Point Down 3106** in Down, Northern Ireland. With an R_final of 0.552 (High band, 93th fleet percentile), its risk profile is dominated by the T (Transition) component at 671% of its weight ceiling, followed by E (Economic) at 556%. Modifiers collectively shift R_base by 8.5%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that the UK's Digital Strategy / Net Zero Strategy planning requires.

CYBER HIGH EXPOSURE

1790
56.8% of fleet

BLIND SPOTS

291
High/Critical + R7 < median

AVG R7 MODIFIER

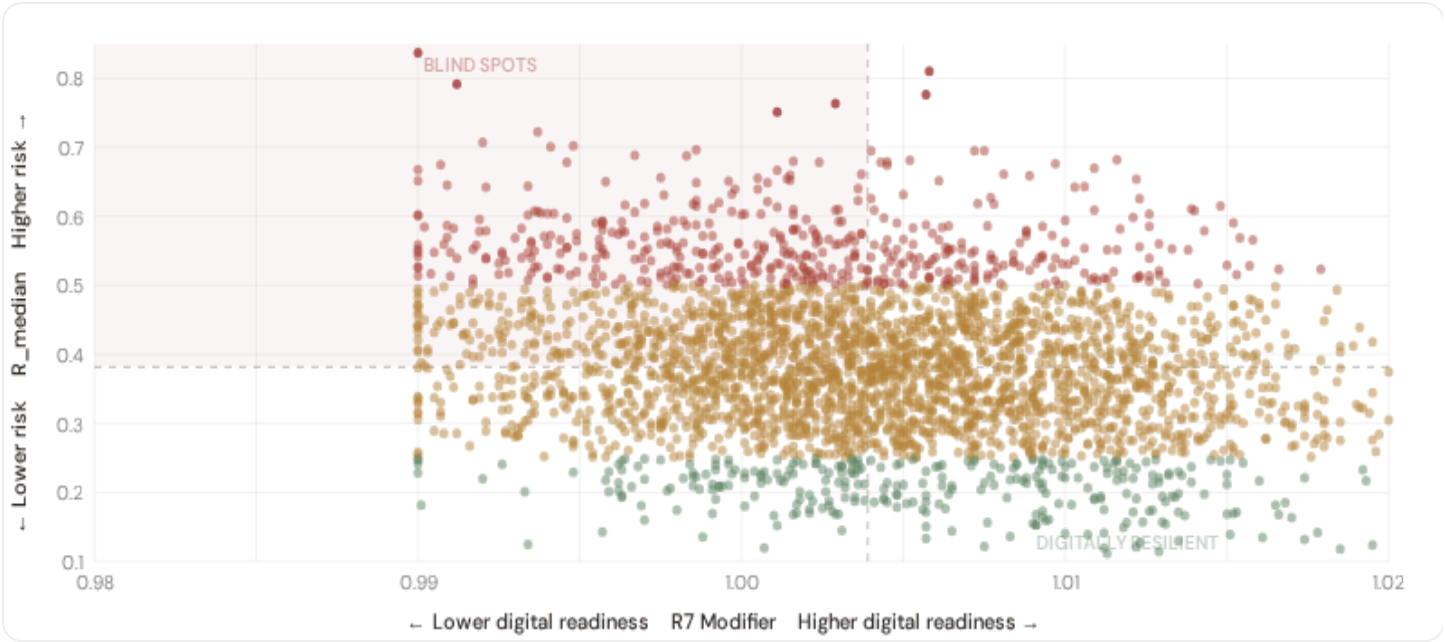
1.0041
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

787
25.0% · R3 ≥ 1.04

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

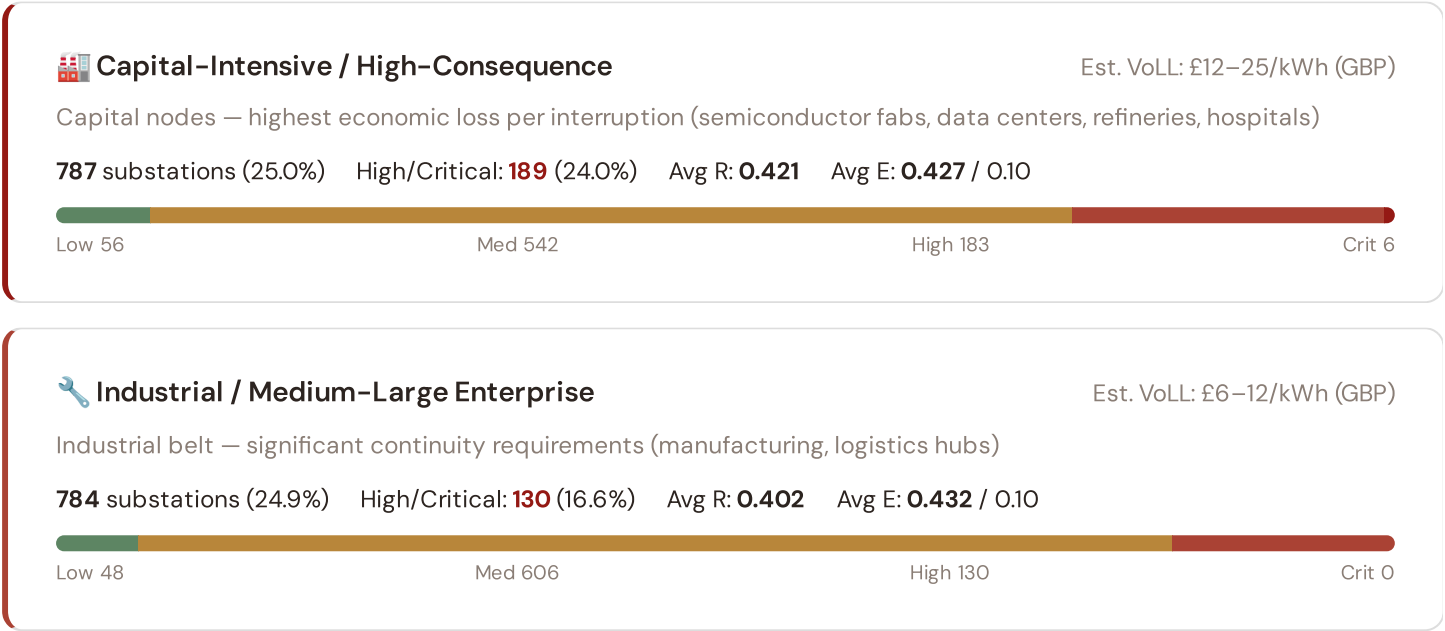


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **291 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0039$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Wales (78), North East (39), North West (36)**. Across the full fleet, 1790 substations (56.8%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Commercial / SME-Dense

Est. VoLL: £3–6/kWh (GBP)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

784 substations (24.9%) High/Critical: **100** (12.8%) Avg R: **0.378** Avg E: **0.431** / 0.10



Light / Agricultural / Rural

Est. VoLL: £1–3/kWh (GBP)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

795 substations (25.2%) High/Critical: **47** (5.9%) Avg R: **0.343** Avg E: **0.403** / 0.10

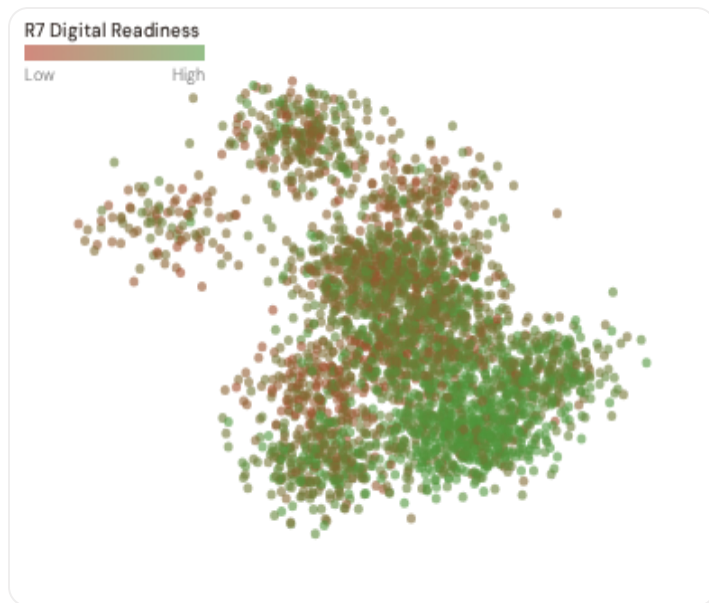


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **189 substations** combine capital-intensive economic fabric ($R3 \geq 1.04$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **North West (99)**, **Yorkshire (80)**, **South West (76)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 14.5%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 County Digital Readiness Ranking

Countys ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify counties combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Dyfed ▲ high R		0.9947
2	Powys ▲ high R		0.9951
3	South Glamorgan ▲ high R		0.9958
4	Gwent ▲ high R		0.9959
5	Clwyd ▲ high R		0.9961
6	Gwynedd ▲ high R		0.9963
7	Antrim ▲ high R		0.9967
8	Down ▲ high R		0.9971
9	Fermanagh ▲ high R		0.9976
10	Cleveland ▲ high R		0.9984
11	Armagh ▲ high R		0.9985
12	Northumberland ▲ high R		0.9989
13	Londonderry ▲ high R		0.9991
14	County Durham ▲ high R		0.9992
15	Tyne and Wear ▲ high R		0.9993

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Dyfed** has the lowest average R7 (0.9947), while **Outer London** leads at 1.0136 — a spread of 0.0189 across the R7 range. **33 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0041, median = 1.0039; 1790 substations classified HIGH cyber-exposure; 291 blind-spot substations (High/Critical risk + below-median R7); 4 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

30
95 variables

WEEKLY / MONTHLY

4
High-frequency feeds

QUARTERLY / ANNUAL

18
Regular refresh cycle

STATIC / DERIVED

8

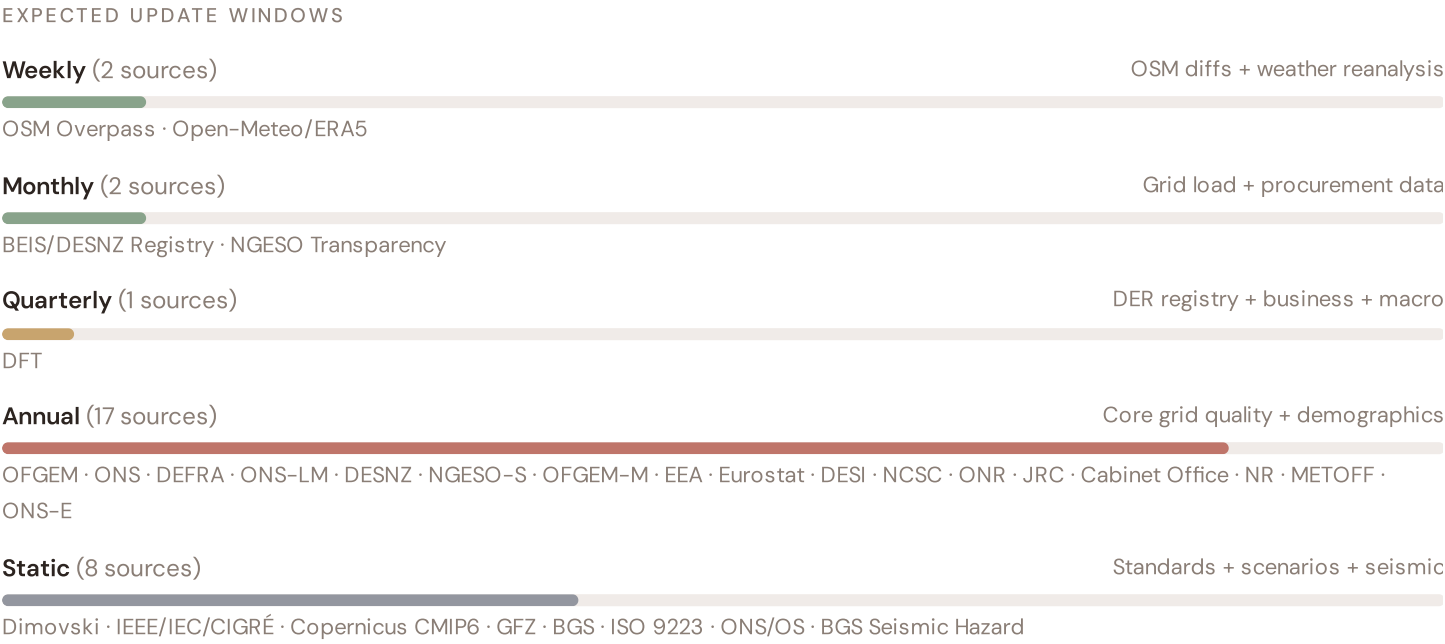
Standards + scenarios

Data Source Registry — June 2026

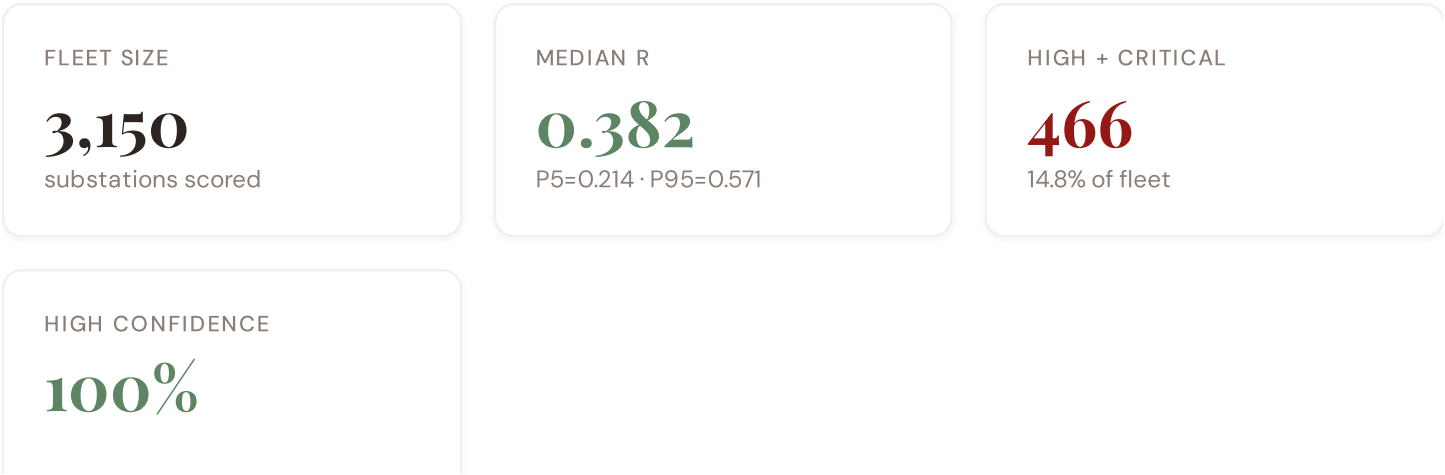
OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
ONS-LM	ONS Labour Market Statistics	MONTHLY	County	1 var
DESNZ	BEIS/DESNZ (Dept for Energy Security and Net Zero)	QUARTERLY	County	5 vars
OFGEM	Ofgem (Office of Gas and Electricity Markets)	ANNUAL	County	8 vars
ONS	ONS (Office for National Statistics)	ANNUAL	County	8 vars
DEFRA	DEFRA/EA (Dept for Environment, Food & Rural Affairs)	ANNUAL	regions	4 vars
JRC	JRC DNO Observatory	ANNUAL	DNO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
NCSC	NCSC (National Cyber Security Centre)	ANNUAL	National	2 vars
EA-FR	EA Flood Risk / Cabinet Office CCA	ANNUAL	County	2 vars
ONR	ONR (Office for Nuclear Regulation)	ANNUAL	Site-level	1 var
ONS-E	ONS Economic Statistics	ANNUAL	regions	3 vars
NGESO-S	NGESO Statistical Series	ANNUAL	regions	2 vars
NR	Network Rail Statistics	ANNUAL	County	1 var
DfT	DfT/DVLA (Driver and Vehicle Licensing Agency)	ANNUAL	County	1 var
OFGEM-M	Ofgem RIIO Monitoring Report	ANNUAL	DNO-level	2 vars
BGS	BGS (British Geological Survey)	STATIC	County	3 vars
CDS	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars

GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
BGS-SH	BGS Seismic Hazard Assessment	STATIC	Municipal	1 var
ONS-GE0	ONS Geography Registry	STATIC	County	1 var
ISO9223	ISO 9223 Corrosion	DERIVED	County	1 var
NGESO	NGESO (National Grid ESO) — Data Portal	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
METOFF	Met Office (UK Met Office)	HOURLY	Station	3 vars

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot



BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **3,150 substation scores remain unchanged** this edition. The fleet median R of 0.382 (mean 0.386) reflects a distribution skewed toward the lower bands, with 9.7% in Low and 75.5% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

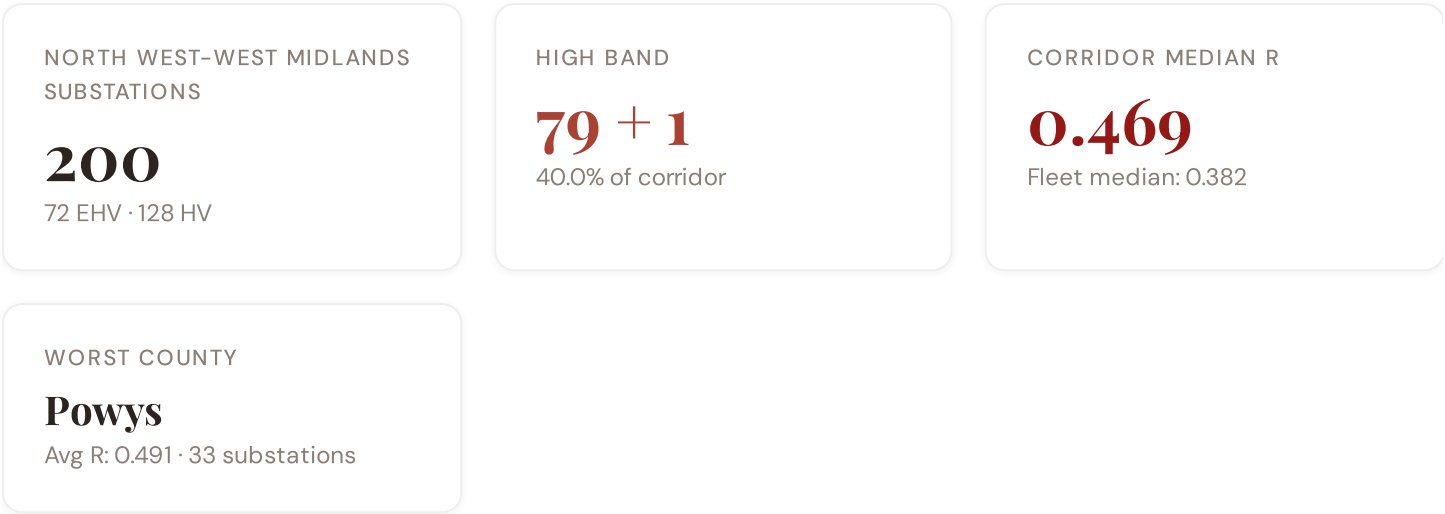
17 changes tracked across PO, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	\$2, \$4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	\$5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	\$5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	\$5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	\$5
F6	ENHANCED	R7 Digital Readiness — County-level DESI model	\$5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	\$2, \$8
L2	ENHANCED	E2 Innovation Enrichment — R&D + economic structure	\$6
L3	ENHANCED	V_socio Fiscal Enrichment — ONS + DEFRA/EA + energy price	\$5
L4	ENHANCED	R3 Demographic Shift Amplifier — ONS net migration	\$5
L5	DATA	95/95 variables operational (100%). 30 data sources total.	\$8, \$12
G1	DATA	BEIS/DESNZ upgraded to quarterly registry — County-level DER registry	\$12
G2	DATA	BGS upgraded to live API — County-level seismic data	\$12
G3	DATA	OSM upgraded to Overpass API — 3,150 real substations	\$12
G4	NEW	R6b Seismic Hazard modifier — BGS Seismic Hazard classification from BGS	\$5
G5	NEW	Network & Topology layer (H): 7 variables — NGESO Grid Plan, seismic analysis	\$12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	\$12

The North West–West Midlands Seismic Corridor: Where Seismic Risk Meets Energy Transition Pressure

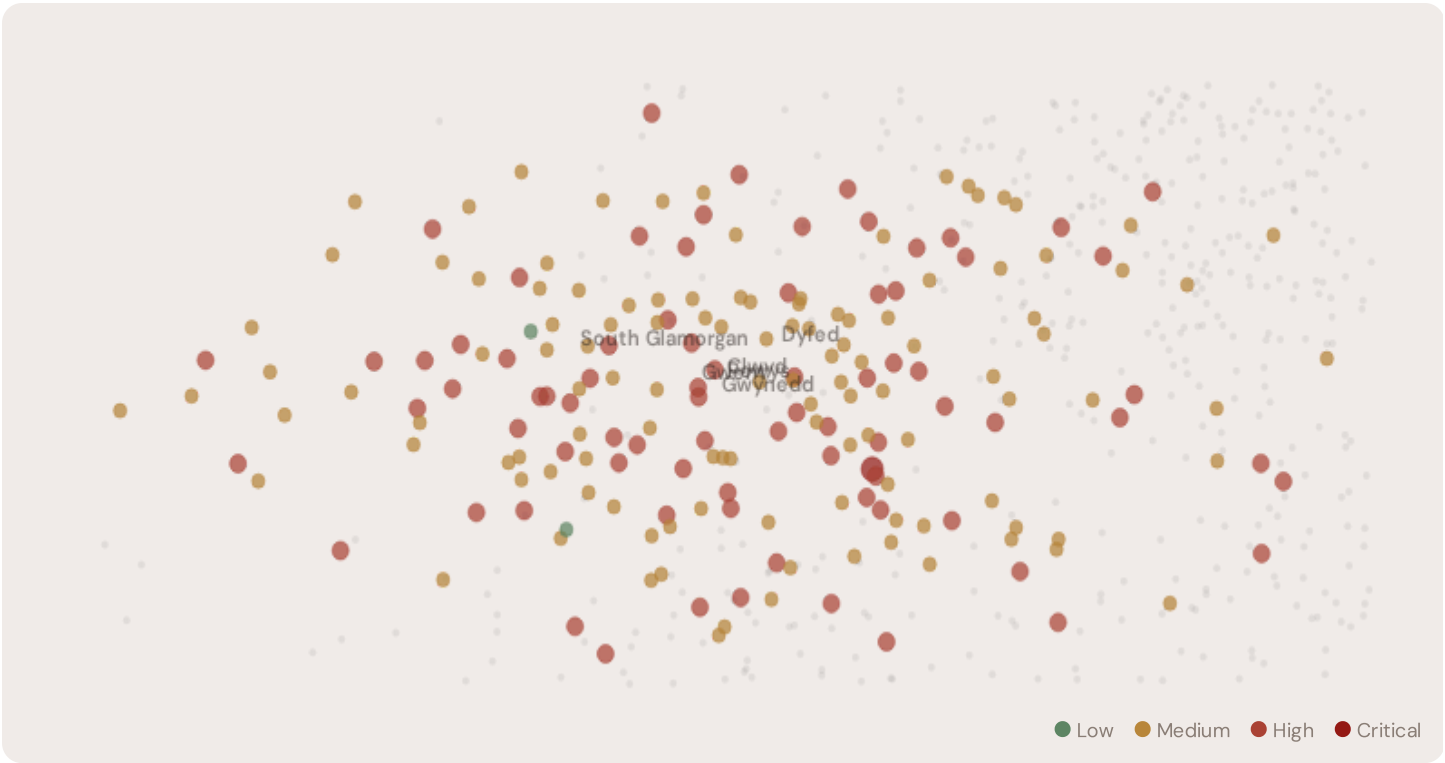
Deep–dive rotation: Each edition features a substantive thematic analysis. The 12–month rotation: **Ed. 001** North West–West Midlands seismic corridor · **002** DER saturation hotspots (East Anglia/West Midlands) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Nuclear fleet dependency and phase–out risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T–component forward projections · **009** County Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the North West–West Midlands, Why Now



The Wales hosts **200 substations** across 6 region regions, making it the locus of this edition's deep–dive analysis. Its risk profile is concentrated in the upper bands: **79 substations (39.5%)** are classified High and **1** Critical, with only **2** in the Low band. 80% of corridor substations score above the national fleet median of 0.382. The corridor median R of 0.469 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity–of–supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	COUNTY	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Primary Gwynedd 2743	Gwynedd	0.837	Critical	0.484	0.707	0.791	0.876	0.958	0.974	C,I
Grid Supply Point Powys 2771	Powys	0.722	High	0.825	0.592	0.707	0.848	0.655	0.509	C,I
Primary South Glamorgan 2733	South Glamorgan	0.707	High	0.548	0.628	0.503	0.615	0.667	0.760	C,I
Substation Powys 2705	Powys	0.702	High	0.772	0.909	0.616	0.817	0.723	0.797	C,I
Primary Gwent 2728	Gwent	0.697	High	0.881	0.568	0.819	0.537	0.671	0.575	C,I
Substation Gwent 2680	Gwent	0.689	High	0.976	0.990	0.401	0.646	0.791	0.534	C,I
Substation Gwynedd 2785	Gwynedd	0.678	High	0.831	0.653	0.684	0.625	0.649	0.613	C,I
BSP Gwent 2722	Gwent	0.668	High	0.770	0.830	0.634	0.813	0.663	0.827	C,I

SUBSTATION	COUNTY	R_FINAL	BAND	C	V	I	E	S	T	ALERT
GSP Gwynedd 2749	Gwynedd	0.650	High	0.700	0.869	0.760	0.607	0.578	0.462	C,I
BSP Clwyd 2732	Clwyd	0.648	High	0.742	0.660	0.673	0.611	0.879	0.903	C,I

... 190 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — WALES VS FLEET

C — Continuity	0.494 vs 0.418 (+18%)
I — Infrastructure	0.508 vs 0.418 (+21%)
S — Saturation	0.512 vs 0.424 (+21%)
V — Voltage	0.519 vs 0.427 (+22%)
E — Economic	0.515 vs 0.423 (+22%)
T — Transition	0.510 vs 0.426 (+20%)

MODIFIER AVERAGES — WALES VS FLEET

R3 Consequence	1.019	fleet 0.998 ↑
R4 Graph Crit.	0.921	fleet 0.916 ↓
R6a Restoration	0.999	fleet 0.999 →
R6b Seismic	1.003	fleet 1.001 →
R7 Digital Read.	0.996	fleet 1.004 →

The dominant risk driver across the Wales corridor is **T (Transition)**, averaging 0.510 against a fleet average of 0.426 — occupying 1020% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.519 vs fleet 0.427 (519% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.019 (fleet: 0.998), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.003, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 County Breakdown

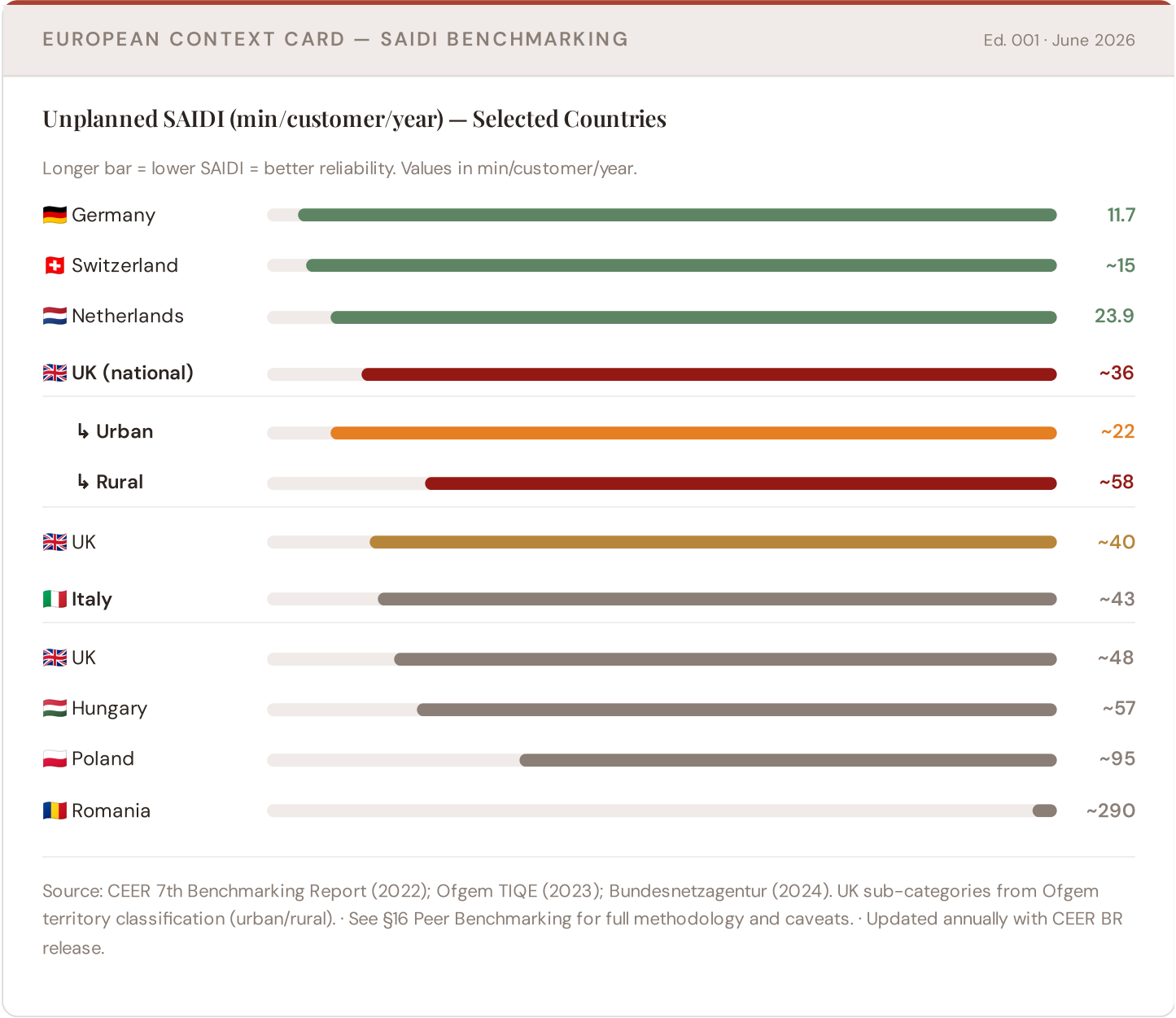
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D.5 Implications

9 Wales substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Powys sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the North West–West Midlands corridor analysis hinges on the UK's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

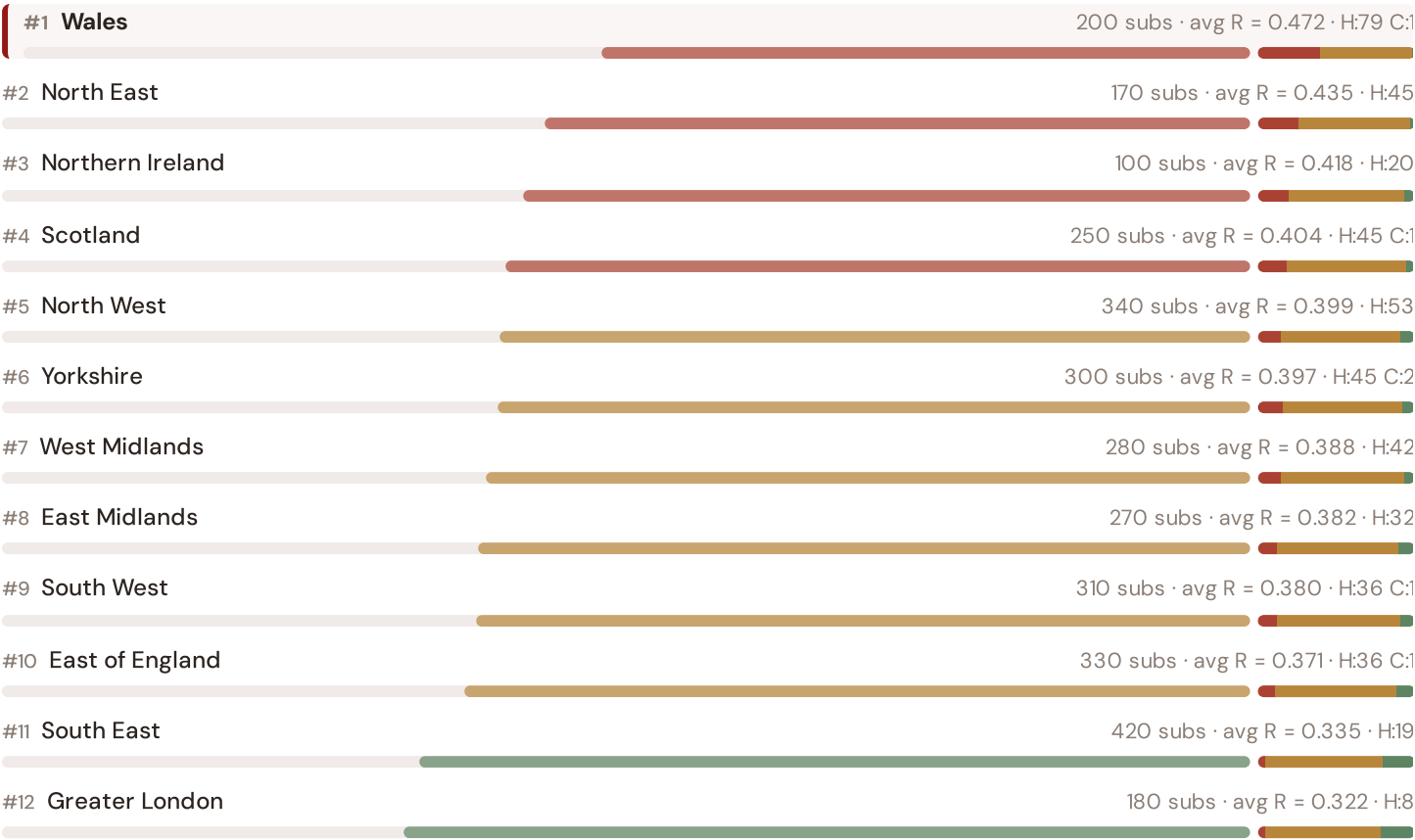


This month's key insight: The Wales corridor deep-dive shows **198 substations** (of 200) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.494 is **1.2× the fleet average** of 0.418. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 17 Regions

Where does the North West–West Midlands sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

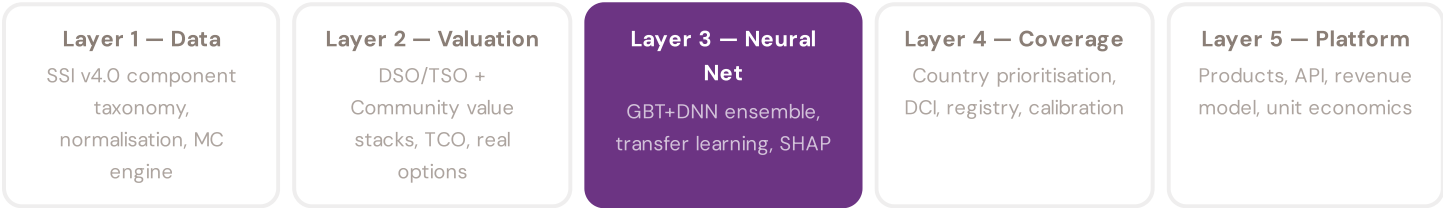


The Wales ranks **#1 of 12 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Wales, North East, Northern Ireland, while the three most resilient are Greater London, South East, East of England. 7 of 12 score above the fleet average of 0.386. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the NGESO/DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 3,150 substations with complete feature vectors. Average CI_width of 0.134 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 306 Low, 2378 Medium, 460 High, 6 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 1 Quarterly Sources

DFT are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Ofgem (Office of Gas and Electricity Markets)** (8 variables → C1–C4 (SAIDI/SAIFI/CAIDI), quality regulation).

FLEET SPOTLIGHT

South Glamorgan: Highest Risk Concentration

South Glamorgan has the highest concentration of High/Critical substations: **17 of 34 (50%)** are in the upper risk bands — avg R = 0.488. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a British utility, Ofgem, NGESO, regional DNO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

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10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5